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Supplemental information

Toripalimab plus chemotherapy in treatment-naïve, advanced esophageal squamous cell carcinoma (JUPITER-06): A multi-center phase 3 trial

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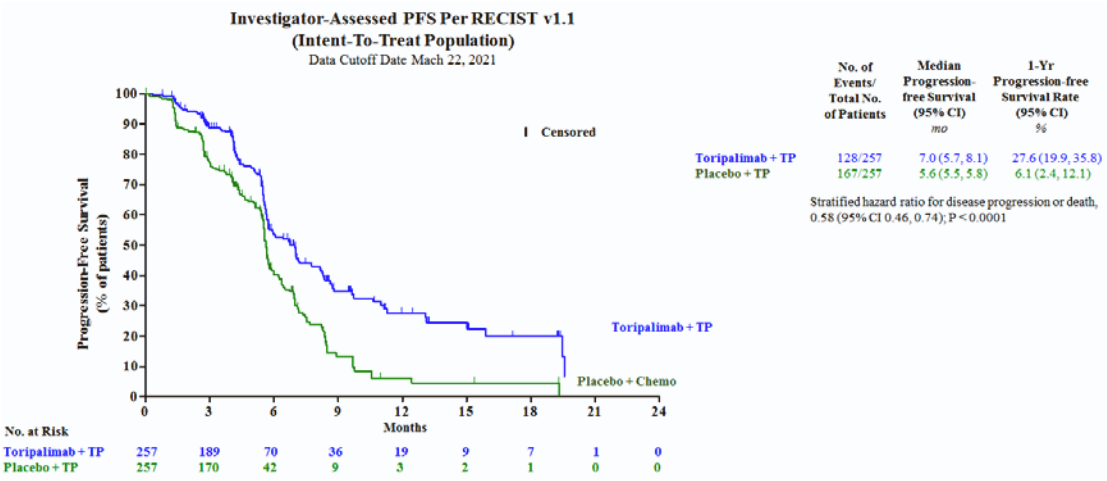
Supplementary Figure 1. Kaplan-Meier-estimated PFS curves as assessed by the investigator according to RECIST v1.1. Censored patients are marked with " ┃ " in the graph. Numbers of patients at risk at indicated time points shown below x-axis. Number of events, median PFS, 1-year PFS rates and stratified hazard ratio for PFS shown to the right of Kaplan-Meier curves (Related to Figure 2).



Supplemental Table 1. TNM classification at enrollment (Related to Table 1).

	Toripalimab + TP (N=257)	Placebo + TP (N=257)	Total (N=514)
TNM Grade primary tumor (T)			
Tx	100 (38.9)	79 (30.7)	179 (34.8)
T0	12 (4.7)	16 (6.2)	28 (5.4)
T1a/T1b	3 (1.2)	4 (1.6)	7 (1.4)
T2	14 (5.4)	18 (7.0)	32 (6.2)
T3	79 (30.7)	86 (33.5)	165 (32.1)
T4a/T4b	47 (18.3)	46 (17.9)	93 (18.1)
Not available	2 (0.8)	8 (3.1)	10 (1.9)
TNM Grade regional lymph node (N)			
Nx	82 (31.9)	59 (23.0)	141 (27.4)
N0	15 (5.8)	10 (3.9)	25 (4.9)
N1	49 (19.1)	53 (20.6)	102 (19.8)
N2	59 (23.0)	70 (27.2)	129 (25.1)
N3	51 (19.8)	61 (23.7)	112 (21.8)
Not available	1 (0.4)	4 (1.6)	5 (1.0)
TNM Grade Metastasis (M)			
M0	51 (19.8)	59 (23.0)	110 (21.4)
M1	206 (80.2)	198 (77.0)	404 (78.6)

Supplemental Table 2. Analysis data sets (Related to Table 1).

	Toripalimab + TP (N=257)	Placebo + TP (N=257)	Total (N=514)
Randomized	257	257	514
Intention-to-treat Analysis Set	257	257	514
Per-Protocol Analysis Set	255 (99.2%)	255 (99.2%)	510 (99.2%)
Safety Analysis Set	257 (100%)	257 (100%)	514 (100%)

Supplemental Table 3. Per-Protocol Analysis Set Exclusions (Related to Table 1).

Subject ID / Arm	Major Protocol Violation
1007011 / toripalimab	Prior systemic anti-tumor therapy (treatment regimen contained docetaxel and nedaplatin), which did not meet the inclusion criterion.
1009002 / toripalimab	A second primary gastric adenocarcinoma, which met an exclusion criterion.
1015014 / Placebo	Identified to have hypopharyngeal carcinoma in situ by pathology report after enrollment, which met an exclusion criterion.
1047002 / Placebo	Uncontrolled tumor related pain before enrollment, which met an exclusion criterion.

Supplemental Table 4. Subsequent Anti-Cancer Therapy in Intent-to-Treat Population After the Study Treatment (Related to Table 1).

Treatment	Toripalimab + TP (N=257) n (%)	Placebo + TP (N=257) n (%)	Total (N=514) n (%)
Patients with at least one subsequent anti-cancer therapy	99 (38.5)	130 (50.6)	229 (44.6)
Chemotherapy	62 (24.1)	76 (29.6)	138 (26.8)
Radiation therapy	30 (11.7)	44 (17.1)	74 (14.4)
Targeting therapy	23 (8.9)	46 (17.9)	69 (13.4)
Anti-PD-1/PD-L1 therapy[1]	17 (6.6)	39 (15.2)	56 (10.9)
Camrelizumab	10 (3.9)	16 (6.2)	26 (5.1)
Toripalimab	4 (1.6)	10 (3.9)	14 (2.7)
Sintilimab	3 (1.2)	9 (3.5)	12 (2.3)
Pembrolizumab	0	3 (1.2)	3 (0.6)
Tislelizumab	0	1 (0.4)	1 (0.2)
Other anti-PD-1	1 (0.4)	0	1 (0.2)
Surgery	4 (1.6)	4 (1.6)	8 (1.6)
Others[2]	26 (10.1)	31 (12.1)	57 (11.1)

[1] One patient from the toripalimab arm received two anti-PD-1therapies after discontinuation from the study treatment.

[2] Others: includes traditional oriental herbal therapy, photodynamic therapy and other unclassified anti-tumor therapy.

Supplemental Table 5. Progression-free survival evaluated by BICR in accordance with RECIST v1.1 (BICR-PFS) in the Intent-to-Treat Population (Related to Figure 2).

	Toripalimab + TP (N=257) n (%)	Placebo + TP (N=257) n (%)
Number of PFS events	132 (51.4)	164 (63.8)
Disease progression	117 (45.5)	152 (59.1)
Death	15 (5.8)	12 (4.7)
Number of censored patients	125 (48.6)	93 (36.2)
Without death and no tumor assessment after randomization	4 (1.6)	3 (1.2)
Missing baseline tumor evaluation	0	1 (0.4)
PD or death after missing ≥2 consecutive tumor evaluations	6 (2.3)	16 (6.2)
PD or death after subsequent anti-tumor treatment	2 (0.8)	4 (1.6)
No PD, no death, and started subsequent anti-tumor treatment	19 (7.4)	10 (3.9)
No PD, no death, and no subsequent anti-tumor treatment	94 (36.6)	59 (23.0)
Progression-free survival (months)		
25% quantile (95% CI)	4.2 (4.0, 5.4)	3.1 (2.7, 4.1)
Median (95% CI)	5.7 (5.6, 7.0)	5.5 (5.2, 5.6)
75% quantile (95% CI)	12.4 (9.7, NE)	7.0 (5.8, 7.5)
Minimum, maximum	0.033+, 21.355+	0.033+, 19.285+
12-month PFS rate (%) (95% CI)	27.8 (20.4, 35.8)	6.1 (2.2, 12.6)
24-month PFS rate (%) (95% CI)	NE (NE, NE)	NE (NE, NE)
Toripalimab + TP vs. placebo + TP		
Stratified analysis		
Hazard ratio (95% CI)	0.58 (0.461, 0.738)	
p-value	<0.0001	
Non-stratified analysis		
Hazard ratio (95% CI)	0.57 (0.451, 0.719)	
p-value	<0.0001	

CI, confidence interval, NE, not estimable.

Note: BIRC assessed PFS is defined as the time from randomization until the earliest occurrence of disease progression, as determined by BIRC per RECIST v1.1, or death from any cause, whichever occurs first.

Note: PFS (in month) = (event date or censoring date - randomization date + 1)/30.4375.

HR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were: ECOG

PS score (0 vs. 1), prior radiotherapy (yes vs. no).

P-value (two-sided) was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 6. Progression-free survival evaluated by BICR in accordance with RECIST v1.1 (BICR-PFS) in the Per-protocol Population (Related to Figure 2)

	Toripalimab + TP (N=255) n (%)	Placebo + TP (N=255) n (%)
Number of PFS events	130 (51.0)	162 (63.5)
Disease progression	116 (45.5)	151 (59.2)
Death	14 (5.5)	11 (4.3)
Number of censored patients	125 (48.6)	93 (36.2)
Without death and no tumor assessment after randomization	4 (1.6)	3 (1.2)
Missing baseline tumor evaluation	0	1 (0.4)
PD or death after missing ≥2 consecutive tumor evaluations	6 (2.3)	16 (6.2)
PD or death after subsequent anti-tumor treatment	2 (0.8)	4 (1.6)
No PD, no death, and starting subsequent anti-tumor treatment	19 (7.4)	10 (3.9)
No PD, no death, and no subsequent anti-tumor treatment	94 (36.6)	59 (23.0)
Progression-free survival (months)		
25% quantile (95% CI)	4.2 (4.0, 5.4)	3.7 (2.7, 4.1)
Median (95% CI)	5.7 (5.6, 7.0)	5.5 (5.3, 5.6)
75% quantile (95% CI)	13.1 (9.7, NE)	7.0 (5.8, 7.5)
Minimum, maximum	0.033+, 21.355+	0.033+, 19.285+
12-month PFS rate (%) (95% CI)	28.3 (20.8, 36.3)	6.1 (2.3, 12.7)
24-month PFS rate (%) (95% CI)	NE (NE, NE)	NE (NE, NE)
Toripalimab + TP vs. placebo + TP		
Stratified analysis		
Hazard ratio (95% CI)	0.59 (0.463, 0.744)	
p-value	<0.0001	
Non-stratified analysis		
Hazard ratio (95% CI)	0.57 (0.453, 0.724)	
p-value	<0.0001	

CI, confidence interval, NE, not estimable.

Note: BIRC assessed PFS is defined as the time from randomization until the earliest occurrence of disease progression, as determined by BIRC per RECIST v1.1, or death from any cause, whichever occurs first.

Note: PFS (in month) = (event date or censoring date - randomization date + 1)/30.4375.

HR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were: ECOG

PS score (0 vs. 1), prior radiotherapy (yes vs. no).

P-value (two-sided) was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 7. Progression-free survival evaluated by the investigator in accordance with RECIST 1.1 (INV-PFS) in the Intent-to-Treat Population (Related to Figure 2)

	Toripalimab + TP (N=257) n (%)	Placebo + TP (N=257) n (%)
Number of PFS events	128 (49.8)	167 (65.0)
Disease progression	109 (42.4)	149 (58.0)
Death between two tumor evaluations	19 (7.4)	18 (7.0)
Number of censored patients	129 (50.2)	90 (35.0)
Without death, and no tumor assessment after randomization	4 (1.6)	3 (1.2)
Missing baseline tumor evaluation	0	0
PD or death after missing ≥2 consecutive tumor evaluations	7 (2.7)	9 (3.5)
PD or death after subsequent anti-tumor treatment	0	1 (0.4)
No PD, no death, and starting subsequent anti-tumor treatment	14 (5.4)	5 (1.9)
No PD, no death, and no subsequent anti-tumor treatment	104 (40.5)	72 (28.0)
Progression-free survival (months)		
25% quantile (95% CI)	5.0 (4.2, 5.5)	3.4 (2.8, 4.1)
Median (95% CI)	7.0 (5.7, 8.1)	5.6 (5.5, 5.8)
75% quantile (95% CI)	13.1 (9.7, 19.5)	7.5 (7.0, 8.4)
Minimum, maximum	0.033+, 19.548+	0.033+, 19.318
12-month PFS rate (%) (95% CI)	27.6 (19.9, 35.8)	6.1 (2.4, 12.1)
24-month PFS rate (%) (95% CI)	NE (NE, NE)	0 (0, 0)
Toripalimab + TP vs. placebo + TP		
Stratified analysis		
Hazard ratio (95% CI)	0.58 (0.457, 0.735)	
p-value	<0.0001	
Non-stratified analysis		
Hazard ratio (95% CI)	0.57 (0.449, 0.718)	
p-value	<0.0001	

CI, confidence interval, NE, not estimable.

Note: Investigator assessed PFS is defined as the time from randomization until the earliest occurrence of disease progression, as determined by the investigator per RECIST v1.1, or death from any cause, whichever occurs first.

Note: PFS (in month) = (event date or censoring date - randomization date + 1)/30.4375.

HR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were: ECOG

PS score (0 vs. 1), prior radiotherapy (yes vs. no).

P-value (two-sided) was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 8. Progression-free survival evaluated by the investigator in accordance with RECIST v1.1 (INV-PFS) in the Per-protocol Population (Related to Figure 2)

	Toripalimab + TP (N=255) n (%)	Placebo + TP (N=255) n (%)
Number of PFS events	127 (49.8)	165 (64.7)
Disease progression	109 (42.7)	147 (57.6)
Death between two tumor evaluations	18 (7.0)	18 (7.0)
Number of censored patients	129 (50.2)	90 (35.0)
Without death, and no tumor assessment after randomization	4 (1.6)	3 (1.2)
Missing baseline tumor evaluation	0	0
PD or death after missing ≥2 consecutive tumor evaluations	7 (2.7)	9 (3.5)
PD or death after subsequent anti-tumor treatment	0	1 (0.4)
No PD, no death, and starting subsequent anti-tumor treatment	14 (5.4)	5 (1.9)
No PD, no death, and no subsequent anti-tumor treatment	104 (40.5)	72 (28.0)
Progression-free survival (months)		
25% quantile (95% CI)	5.0 (4.2, 5.5)	3.7 (2.8, 4.3)
Median (95% CI)	6.8 (5.7, 8.1)	5.6 (5.5, 5.8)
75% quantile (95% CI)	13.1 (10.6, 19.5)	7.5 (7.0, 8.4)
Minimum, maximum	0.033+, 19.548+	0.033+, 19.318
12-month PFS rate (%) (95% CI)	27.9 (20.2, 36.2)	6.1 (2.5, 12.2)
24-month PFS rate (%) (95% CI)	NE (NE, NE)	0 (0, 0)
Toripalimab + TP vs. placebo + TP		
Stratified analysis		
Hazard ratio (95% CI)	0.59 (0.464, 0.749)	
p-value	<0.0001	
Non-stratified analysis		
Hazard ratio (95% CI)	0.58 (0.455, 0.730)	
p-value	<0.0001	

CI, confidence interval, NE, not estimable.

Note: Investigator assessed PFS is defined as the time from randomization until the earliest occurrence of disease progression, as determined by the investigator per RECIST v1.1, or death from any cause, whichever occurs first.

Note: PFS (in month) = (event date or censoring date - randomization date + 1)/30.4375.

HR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were: ECOG

PS score (0 vs. 1), prior radiotherapy (yes vs. no).

P-value (two-sided) was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 9. Overall survival in the Intent-to-Treat population (Related to Figure 3).

	Toripalimab + TP (N=257) n (%)	Placebo + TP (N=257) n (%)
Number of deaths	70 (27.2)	103 (40.1)
Number of censored patients	187 (72.8)	154 (59.9)
Overall survival (months)		
25% quantile (95% CI)	8.1 (7.3, 10.9)	7.4 (6.9, 8.2)
Median (95% CI)	17.0 (14.0, NE)	11.0 (10.4, 12.6)
75% quantile (95% CI)	NE (NE, NE)	18.3 (13.4, NE)
Minimum, maximum	0.329, 23.918+	0.164, 24.641+
12-month survival rate (%) (95% CI)	66.0 (57.5, 73.2)	43.7 (34.4, 52.6)
24-month survival rate (%) (95% CI)	NE (NE, NE)	17.5 (8.7, 28.9)
Toripalimab + TP vs. placebo + TP		
Stratified analysis		
Hazard ratio (95% CI)	0.58 (0.425, 0.783)	
p-value	0.0004	
Non-stratified analysis		
Hazard ratio (95% CI)	0.59 (0.433, 0.797)	
p-value	0.0006	

CI, confidence interval, NE, not estimable.

Note: OS is defined as the time from randomization to death from any cause and is calculated as (death date or censoring date - randomization date + 1)/30.4375 in month.

HR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were: ECOG PS score (0 vs. 1), prior radiotherapy (yes vs. no).

P-value (two-sided) was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 10. Overall survival in the Per-protocol population (Related to Figure 3).

	Toripalimab + TP (N=255) n (%)	Placebo + TP (N=255) n (%)
Number of deaths	69 (27.1)	101 (39.6)
Number of censored patients	186 (72.9)	154 (60.4)
Overall survival (months)		
25% quantile (95% CI)	8.1 (7.3, 10.9)	7.5 (6.9, 8.2)
Median (95% CI)	17.0 (14.0, NE)	11.1 (10.4, 12.6)
75% quantile (95% CI)	NE (NE, NE)	18.3 (13.4, NE)
Minimum, maximum	0.329, 23.918+	0.164, 24.641+
12-month survival rate (%) (95% CI)	66.5 (58.1, 73.7)	44.1 (34.7, 53.0)
24-month survival rate (%) (95% CI)	NE (NE, NE)	17.7 (8.7, 29.2)
Toripalimab + TP vs. placebo + TP		
Stratified analysis		
Hazard ratio (95% CI)	0.58 (0.429, 0.795)	
p-value	0.0006	
Non-stratified analysis		
Hazard ratio (95% CI)	0.59 (0.435, 0.805)	
p-value	0.0008	

CI, confidence interval, NE, not estimable.

Note: OS is defined as the time from randomization to death from any cause and is calculated as (death date or censoring date - randomization date + 1)/30.4375 in month.

HR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were: ECOG PS score (0 vs. 1), prior radiotherapy (yes vs. no).

P-value (two-sided) was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 11. Objective response rate (INV-ORR), disease control rate (INV-DCR) and duration of response (INV-DoR) evaluated by investigators in accordance with RECIST v1.1 in the Intent-to-Treat Population (Related to Table 2).

Variable	Toripalimab + TP (N=257)	Placebo + TP (N=257)
Best overall response - no. (%)		
Complete response	13 (5.1)	8 (3.1)
Partial response	173 (67.3)	142 (55.3)
Stable disease	52 (20.2)	74 (28.8)
Progressive disease	10 (3.9)	24 (9.3)
Not evaluable	9 (3.5)	9 (3.5)
Objective response rate (ORR)		
ORR % (95% CI)	72.4 (66.5-77.7)	58.4 (52.1-64.5)
Difference in ORR % (95% CI)	14.1 (6.0-22.2)	
P-value[1]	0.0007	
Disease control rate (DCR)		
DCR % (95% CI)	92.6 (88.7-95.5)	87.2 (82.4-91.0)
Difference in DCR % (95% CI)	5.5 (0.3-10.7)	
P-value[1]	0.0371	
Estimate of median DoR (95% CI) (months)	7.0 (5.7-8.4)	4.3 (4.2-5.0)
Stratified estimate of hazard ratio (toripalimab versus placebo) (95%CI) [2]	0.50 (0.36-0.69)	
P-value[3]	0.0001	

TP, paclitaxel and cisplatin; CR, complete response; PD, progressive disease; NE, not estimable.

[1]In the stratified analysis, nominal p-value (two-sided) of the comparison between arms was calculated using Cochran-Mantel-Haenszel method based on random stratification factors, and the 95% confidence interval for the difference in the rate difference between the arms was estimated based on Mantel-Haenszel, stratified by ECOG PS score (0 vs. 1), and prior radiotherapy (yes vs. no).

[2]HR for DoR was estimated by the Cox proportional hazard model, and the stratified analysis adopted the stratified Cox proportional hazard model. The stratification factors during the randomization process were: ECOG PS score (0 vs. 1), prior radiotherapy (yes vs. no).

[3]P-value (two-sided) for DoR was calculated using log-rank test, stratified analysis was performed using stratified log-rank test, and the stratification factors were the same as HR.

Supplemental Table 12. Study Drug Exposure (Related to Table 3).

	Toripalimab +TP (N=257)	Placebo +TP (N=257)
The number of toripalimab/placebo cycles		
Mean	8.3	7.1
Sd	5.7	4.1
Median	7.0	7.0
Minimum, Maximum	1, 33	1, 26
Weeks of exposure for toripalimab/placebo		
Mean	23.8	19.7
Sd	17.8	13.0
Median	21.0	19.4
Minimum, Maximum	0.1, 96.1	0.1, 84.3
Cumulative dose for toripalimab/placebo (mg)		
Mean	1993	1709
Sd	1366	978
Median	1680	1680
Minimum, Maximum	240, 7920	240, 6240
Relative dose intensity for toripalimab/placebo (%)		
Mean	92.6	94.2
Sd	9.2	7.6
Median	95.6	96.6
Minimum, Maximum	36.5, 101.9	52.2, 102.5
The number of cycles cisplatin		
Mean	5.2	5.1
Sd	2.2	2.5
Median	6.0	6.0
Minimum, Maximum	1, 18	1, 18
Weeks of exposure for cisplatin		
Mean	12.9	12.2
Sd	5.4	5.5
Median	15.1	15.0
Minimum, Maximum	0.2, 27.9	0.2, 25.7
Cumulative dose for cisplatin (mg)		
Mean	569	554
Sd	207	207
Median	630	594
Minimum, Maximum	96, 876	80, 939

	Toripalimab +TP (N=257)	Placebo +TP (N=257)
Relative dose intensity for cisplatin (%)		
Mean	95.2	96.9
Sd	37.3	38.1
Median	92.7	94.2
Minimum, Maximum	36.1, 308.2	40.1, 359.5
The number of cycles paclitaxel		
Mean	5.0	4.8
Sd	1.5	1.6
Median	6.0	6.0
Minimum, Maximum	0, 6	1, 6
Weeks of exposure for paclitaxel		
Mean	13.3	12.5
Sd	5.1	5.3
Median	15.2	15.0
Minimum, Maximum	0.0, 27.9	0.1, 25.7
Cumulative dose for paclitaxel (mg)		
Mean	1382	1345
Sd	469	481
Median	1553	1471
Minimum, Maximum	0, 2043	28, 2192
Relative dose intensity for paclitaxel (%)		
Mean	88.6	90.8
Sd	11.4	11.3
Median	92.8	94.8
Minimum, Maximum	36.0, 109.8	9.1, 105.0

Sd, standard deviation.

Note: Actual dose intensity (ADI) (mg/week） = Average Actual Cumulative Dose received per week.

Note: Relative dose intensity (RDI) (%) = (Actual dose intensity (ADI) / Planned dose intensity) *100%, and Planned dose intensity is defined as first (first treatment cycle) planned dose divided by the number of weeks of 1 planned treatment cycle (e.g. 3 weeks).

Supplemental Table 13. Summary of Treatment Emergent Adverse Events (Safety Population. Related to Table 3)

Adverse Event Category	**Toripalimab + TP** (N=257) n (%)	**Placebo + TP** (N=257) n (%)
Any TEAE	255 (99.2)	255 (99.2)
Any TEAE related to study treatment	250 (97.3)	250 (97.3)
Any TEAE related to toripalimab/placebo	183 (71.2)	158 (61.5)
Any TEAE Grade ≥3	188 (73.2)	180 (70.0)
Any TEAE Grade ≥3 related to study treatment	166 (64.6)	144 (56.0)
Any TEAE Grade ≥3 related to toripalimab/placebo	61 (23.7)	34 (13.2)
Any TEAEs resulting in death	21 (8.2)	21 (8.2)
Any TEAEs resulting in death related to study treatment	1 (0.4)	3 (1.2)
Any TEAEs resulting in death related to toripalimab/placebo	1 (0.4)	2 (0.8)
Any SAE	93 (36.2)	74 (28.8)
Any SAE related to study treatment	60 (23.3)	38 (14.8)
Any SAE related to toripalimab/placebo	37 (14.4)	14 (5.4)
Any TEAE leading to discontinuation of toripalimab/placebo	30 (11.7)	16 (6.2)
Any TEAE leading to interruption of toripalimab/placebo	92 (35.8)	66 (25.7)
Any irAE [1]	95 (37.0)	68 (26.5)
Any irAE with Grade ≥3	18 (7.0)	4 (1.6)
Any IRR	9 (3.5)	8 (3.1)

TEAE: treatment-emergent adverse event, SAE: serious adverse event, irAE: immune-related adverse event, IRR: infusion-related reaction.

Note: TEAEs are adverse events that either start or worsen in severity on or after the date/time of first dose of study treatment and on or before 60 days after the date/time of last dose of study treatment or before starting a new anti-cancer therapy, whichever occurs first.

[1] As assessed by the Investigator

Data Cutoff: March 22, 2021

Supplemental Table 14. Immune related adverse events (irAE) reported in more than one patient in either arm as assessed by the investigator (Related to Table 3).

No. of patients (%)	Toripalimab + TP (N=257)		Placebo + TP (N=257)	
	All Grades	Grades ≥3	All Grades	Grade ≥3
Any irAE	95 (37.0%)	18 (7.0%)	68 (26.5%)	4 (1.6%)
Rash	26 (10.1%)	5 (1.9%)	9 (3.5%)	0
Hypothyroidism	15 (5.8%)	0	9 (3.5%)	0
Pruritus	11 (4.3%)	1 (0.4%)	4 (1.6%)	0
Fatigue	10 (3.9%)	1 (0.4%)	9 (3.5%)	0
Hyperglycemia	10 (3.9%)	0	7 (2.7%)	0
Thyroid function test abnormal	10 (3.9%)	0	5 (1.9%)	0
Alanine aminotransferase increased	7 (2.7%)	1 (0.4%)	4 (1.6%)	0
Aspartate aminotransferase increased	6 (2.3%)	1 (0.4%)	3 (1.2%)	0
Diarrhea	6 (2.3%)	0	2 (0.8%)	0
Neutropenia	5 (1.9%)	1 (0.4%)	3 (1.2%)	0
Pneumonitis	5 (1.9%)	0	2 (0.8%)	0
Anemia	4 (1.6%)	0	6 (2.3%)	1 (0.4%)
Arrhythmia	4 (1.6%)	0	3 (1.2%)	0
Decreased appetite	4 (1.6%)	0	4 (1.6%)	0
Hepatic function abnormal	4 (1.6%)	1 (0.4%)	0	0
Leukopenia	4 (1.6%)	0	1 (0.4%)	0
Nausea	4 (1.6%)	0	7 (2.7%)	0
Blood creatine phosphokinase increased	3 (1.2%)	0	0	0
Constipation	3 (1.2%)	0	3 (1.2%)	0
Hyperthyroidism	3 (1.2%)	0	0	0
Thrombocytopenia	3 (1.2%)	0	1 (0.4%)	1 (0.4%)
Vomiting	3 (1.2%)	0	3 (1.2%)	0
Blood urea increased	2 (0.8%)	0	0	0
Enteritis	2 (0.8%)	2 (0.8%)	0	0
Gamma-glutamyl transferase increased	2 (0.8%)	1 (0.4%)	0	0
Heart injury	3 (1.2%)	1 (0.4%)	1 (0.4%)	0
Hypertension	2 (0.8%)	1 (0.4%)	0	0
Immune-mediated myocarditis	1 (0.4%)	1 (0.4%)	0	0
Pyrexia	2 (0.8%)	0	2 (0.8%)	0
Immune-mediated hepatitis	1 (0.4%)	1 (0.4%)	2 (0.8%)	1 (0.4%)
Hyperlipidemia	1 (0.4%)	0	2 (0.8%)	0
Musculoskeletal pain	1 (0.4%)	0	4 (1.6%)	0
Arthritis	0	0	2 (0.8%)	0
Myelosuppression	0	0	3 (1.2%)	1 (0.4%)
Proteinuria	0	0	2 (0.8%)	0

Note: For the same patient if the same adverse event occurs multiple times, that patient is counted only once in the same system organ classification and preferred terminology when calculating the number of adverse events.
Data Cutoff: Mar 22 2021

Supplemental Table 15. Comparability study among different PD-L1 IHC staining antibodies including JS311, 22C3 and 28-8 in tumor biopsies from ESCC patients (n=50). JS311 (CPS≥10) showed 84% and 90% overall concordance with 22C3 (CPS≥10) and 28-8 (TPS≥1) for PD-L1 IHC staining in ESCC (Related to STAR Methods).

ESCC (n=50)		**22C3（CPS≥10）**		**28-8(TPS≥1%)**	
		+	-	+	-
JS311（CPS≥10）	+	1	8	6	3
	-	0	41	2	39